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Systems and methods for electrically heating a catalyst with a honeycomb body having radial walls

Abstract

An electrical heater and a method of heating a catalyst. The heater includes a honeycomb body that includes a honeycomb structure. The honeycomb structure includes a central axis extending longitudinally and a plurality of interconnected walls. The interconnected walls include a plurality of radial walls extending along a radius of the honeycomb body between the central axis and an outermost periphery of the honeycomb body and a plurality of angular walls arranged concentrically with respect to the central axis and spanning between the radial walls. The honeycomb structure includes a plurality of cells defined by the interconnected walls. The heater comprises a first electrode disposed at the central axis and a second electrode disposed radially outwardly of the central axis and in electrical communication with the first electrode via one or more of the intersecting walls that are located between the first electrode and second electrode.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This is a national stage application under 35 U.S.C. § 371 of International Application No. PCT/US2020/049895, filed on Sep. 9, 2020, which claims the benefit of priority under 35 U.S.C § 120 of U.S. Provisional Application Ser. No. 62/900,984 filed on Sep. 16, 2019, the content of which is relied upon and incorporated herein by reference in their entireties.

BACKGROUND

(1) This application generally relates to electrical heaters for heating a catalyst, such as a catalyst on a substrate of a catalytic converter assembly, and specifically, electrical heaters comprising radial honeycomb structures.

SUMMARY

(2) All examples and features mentioned below can be combined in any technically possible way.
(3) In some aspects, an electrical heater is provided. The electrical heater comprises a honeycomb body comprising a honeycomb structure comprising a central axis extending longitudinally therethrough and a plurality of interconnected walls, the interconnected walls comprising a plurality of radial walls, each radial wall extending at least partially along a radius of the honeycomb body between the central axis and an outermost periphery of the honeycomb body; and a plurality of angular walls arranged concentrically with respect to the central axis and spanning between the radial walls; and a plurality of cells defined by the interconnected walls; a first electrode disposed at the central axis; and a second electrode disposed radially outwardly of the

central axis and in electrical communication with the first electrode via one or more of the intersecting walls that are located between the first electrode and second electrode.

(4) In some embodiments, a radial distance between radially consecutive angular walls of the plurality of angular walls decreases in a radial proximity to the outer periphery of the honeycomb body such that a hydraulic diameter of each cell is substantially similar.

(5) In some embodiments, a first group of the plurality of cells is in a first region directly adjacent to the central axis, wherein a second group of the plurality of cells is disposed in a second region radially outwardly of the first region, wherein a first hydraulic diameter of each cell of the first set of the plurality of cells is less than a second hydraulic diameter of each cell of the second set of the plurality of cells.

(6) In some embodiments, each radial wall of a first cell of the plurality of cells is angularly offset with respect to the radial walls of a second cell of the plurality of cells that is radially adjacent to the first cell.

(7) In some embodiments, the honeycomb body comprises a first resistive region separated from a second resistive region by a transitional angular wall, wherein the first resistive region comprises fewer radial walls than the second resistive region, the first resistive region being located nearer to the central axis than the second resistive region and both the first resistive region and the second resistive region being located between the first and second electrodes.

(8) In some embodiments, the first resistive region and second resistive region are bounded by the transitional angular wall and one of the first electrode, the second electrode, or a second transitional angular wall.

(9) In some embodiments, the radial walls of the first resistive region are proportionally thicker than the radial walls of the second resistive region, such that, when carrying current between the first and second electrodes, heat generated by each radial wall of the first resistive region is substantially equivalent to heat generated by each radial wall of second resistive region.

(10) In some embodiments, the radial walls of the first resistive region are proportionally shorter than the radial walls of the second resistive region, such that, when carrying current between the first and second electrodes, heat generated by each radial wall of the first resistive region is substantially equivalent to heat generated by each radial wall of second resistive region.

(11) In some embodiments, the radial walls of the first resistive region are proportionally axially longer than the radial walls of the second resistive region, such that, when carrying current between the first and second electrodes, heat generated by each radial wall of the first resistive region is substantially equivalent to heat generated by each radial wall of second resistive region.

(12) In some embodiments, the radial walls are thicker than the angular walls.

(13) In some embodiments, the second electrode is disposed at an outer skin disposed about the outer periphery of the honeycomb structure.

(14) In some aspects, an exhaust treatment system is provided. The exhaust treatment system comprises an electrical heater according to any of the above and an exhaust treatment device positioned downstream of the electrical heater.

(15) In some embodiments, the exhaust treatment device comprises a substrate loaded with a catalytic material, a particulate filter, or a combination

(16) In some aspects, a method for heating a catalyst is provided. The method comprises the steps of applying an electric potential difference across a first electrode located at a central axis of a honeycomb body and a second electrode located on the honeycomb body radially outwardly from the first electrode, wherein the honeycomb body comprises a plurality of intersecting walls comprising a plurality of radial walls, each radial wall extending at least partially along a radius of the honeycomb body between the central axis and an outermost periphery of the honeycomb body, at least some of the plurality of radial walls angularly spaced from each other about the central axis and a plurality of angular walls arranged concentrically with respect to the central axis and spanning between the radial walls, wherein the electric potential difference creates a flow of

current through the intersecting walls located between the first electrode and the second electrode that generates resistive heating in the intersecting walls.

(17) In some embodiments, a radial dimension of the cells of the plurality of cells decreases for each subsequent radially adjacent cell in radial proximity to the outer periphery of the honeycomb body such that a hydraulic diameter of each cell is substantially similar.

(18) In some embodiments, a first set of the plurality of cells is disposed about the central axis, wherein a second set of the plurality of cells is disposed about the first set, wherein a first hydraulic diameter of each cell of the first set of the plurality of cells is less than a second hydraulic diameter of each cell of the second set of the plurality of cells.

(19) In some embodiments, one or more radial walls of each cell in a first subset of the cells of the plurality of cells are respectively angularly offset with respect to the radial walls of each cell in a second subset of the cells of the plurality of cells that are radially adjacent to the first subset.

(20) In some embodiments, the second subset of cells is separated from the central axis by at least two angular walls of the plurality of angular walls.

(21) In some embodiments, the honeycomb body comprises a first resistive region separated from a second resistive region by a transitional angular wall, wherein the first resistive region comprises fewer radial walls than the second resistive region, the first resistive region being located nearer to the central axis than the second resistive region and both the first resistive region and the second resistive region being located between the first and second electrodes.

(22) In some embodiments, the radial walls of the first resistive region are thicker than the radial walls of the second resistive region, such that the equivalent resistance of the first resistive region and the second resistive region are substantially similar.

(23) In some embodiments, the radial walls of the first resistive region are shorter than the radial walls of the second resistive region, such that, when carrying current between the first and second electrodes, the heat generated by each radial wall of the first resistive region is substantially equivalent to the heat generated by each radial wall of the second resistive region.

(24) In some embodiments, the radial walls of the first resistive region are axially longer than the radial walls of the second resistive region, such that, when carrying current between the first and second electrodes, the heat generated by each radial wall of the first resistive region is substantially equivalent to the heat generated by each radial wall of second resistive region.

(25) In some embodiments, the radial walls are thicker than the angular walls.

(26) In some embodiments, the second electrode is disposed at an outer skin disposed about the outer periphery.

(27) In some embodiments, the catalyst is loaded on the intersecting walls.

(28) In some embodiments, the method further comprises heating a flow of fluid with the heater and heating the catalyst with the flow of fluid.

(29) The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and the drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 depicts a partial cross-sectional view of an electrical heater having a radial honeycomb structure, according to an example.

(2) FIG. 2 depicts a perspective view of an electrical heater, according to an example.

(3) FIG. 3 depicts an equivalent resistance of radial walls of an electrical heater, according to an example.

(4) FIG. 4A depicts a simplified partial cross-sectional view of radial and angular walls of an

electrical heater, according to an example.

(5) FIG. 4B depicts an equivalent resistance of radial walls of an electrical heater, according to an example.

(6) FIG. 5 depicts a simplified partial cross-sectional view of radial and angular walls of an electrical heater, according to an example.

(7) FIG. 6 depicts a simplified axial cross-section view of radial and angular walls of an electrical heater, according to an example,

(8) FIG. 7 depicts a cross-sectional view of an electrical heater having a radial honeycomb structure, according to an example.

(9) FIG. 8 depicts a heating profile of an electrical heater, according to an example.

(10) FIG. 9 depicts a cross-sectional view of an electrical heater having radial staggered cells, according to an example.

(11) FIG. 10 depicts a partial cross-sectional view of an electrical heater having radial staggered cells, according to an example.

(12) FIG. 11 depicts a heating profile of an electrical heater having radial staggered cells, according to an example.

(13) FIG. 12 depicts a cross-sectional view of electrical heater having staggered cells proximate to an outer periphery, according to an example.

(14) FIG. 13 depicts a heating profile of an electrical heater having radial staggered cells and thickened radial walls, according to an example.

(15) FIG. 14 depicts a heating profile of an electrical heater having a peripheral electrode covering only portions of an outer skin, according to an example.

DETAILED DESCRIPTION

(16) Some catalytic converters rely on the heat of engine exhaust to activate the catalyst in order begin treatment of vehicle emissions. However, a significant proportion of total emissions in some vehicles may occur immediately after starting the vehicle, while the catalyst is still cold.

Embodiments disclosed herein pertain to active heating systems and methods that use electrical heating elements to supplement the existing heat of the engine exhaust, thereby enabling vehicle emissions to be further reduced, particularly after cold-start of the vehicle.

(17) One such method for actively heating the exhaust is via an electrically heated catalyst (EHC) system, which raises the catalyst temperature by supplying electric power from a battery, e.g., the battery of the vehicle, to provide heat to the catalytic converter. Electrically heated catalyst designs, however, are affected by issues such as varying degrees of non-uniform heating profiles, which degrade performance and decrease the efficiency in electrical energy usage.

(18) FIG. 1 illustrates a cross-sectional octant of example electrical heater **100** for heating a catalyst. The electrical heater **100** comprises a honeycomb body **101** comprising a radial web honeycomb structure **102**, a central electrode **104** located at a central axis C of the honeycomb structure **102**, and at least one peripheral electrode **106** that is configured to cause a flow of current through an array of interconnected webs or walls comprised by the honeycomb structure **102**. The interconnected walls comprise a plurality of radial walls **108** and a plurality of angular walls **110**, which, together, define a plurality of cells or channels **112** that extend longitudinally through the honeycomb body **101**.

(19) The peripheral electrode **106** can be disposed to cause a flow of current through at least some of the radial walls **108** located between the electrodes **104** and **106** when a potential is applied between the central electrode **104** and the peripheral electrode **106**. The peripheral electrode **106** can thus be disposed at an outer skin **114** (disposed about the outer periphery of the honeycomb structure **102**) or can be embedded beneath the outer skin **114** at a location suitable for inducing current flow through at least a subset of radial webs **108** (e.g., at the outer periphery of the honeycomb structure). The honeycomb structure **102** can be comprised of an electrically conductive material, such as a metal, conductive ceramic, or combination thereof. The conductive

material can have an electrical resistance such that conduction of current, as a result of the application of an electric potential across central electrode **104** and peripheral electrode **106**, generates heat in the walls **108**, **110** of the honeycomb structure **102**. The potential applied to the central electrode **104** and peripheral electrode **106** can be formed by any suitable voltage source, and can, in alternate examples, be a DC voltage (formed, e.g., by a battery) or an AC voltage.

(20) The honeycomb structure **102** of the honeycomb body **101** can be formed, for example, by extrusion of a ceramic forming material through an extrusion die as a green body, which is then cut, dried, and fired into a final ceramic honeycomb body. Alternatively, the honeycomb structure can be formed by an additive machining operation (e.g., such as three-dimensional printing) or subtractive machining operation (e.g., electrical discharge machining, electrochemical machining, etc.). If a ceramic material is selected that is too resistive to provide effective resistive heating, the ceramic material can be doped with a conductor, such as metal, to increase the conductivity of the ceramic. In an alternative example, the honeycomb structure can be formed entirely of a metal, rather than ceramic material, or can be formed from a conductive ceramic material.

(21) In an example, the plurality of radial webs **108** extend radially between central axis C and the outer periphery of honeycomb structure (e.g., to outer skin **114**), and together can define a plurality radial walls **116**. As the plurality of radial webs **108** extend from the central axis C toward the outer periphery, angularly adjacent radial walls **108** will diverge from one another, thereby increasing the angular distance between adjacent walls **108** at the periphery of the honeycomb body **101**. The plurality of angular walls **110** extend concentrically with respect to the central axis C, such that each angular wall **110** is an arc that is equidistant to the central axis along its length. Together define a plurality of angular walls **110** arranged concentrically about the central axis C. At least a subset of the radial walls **108** can terminate at the outer skin **114**, forming an orthogonal joint **120** with an inner surface of outer skin **114**. In an example, some of the radial walls **108** (e.g., radial wall **108a**) can extend almost completely the radial distance between the central axis C of the honeycomb structure **102** and the outer skin **114**, while other radial walls **108** (e.g., radial wall **108b**) extend only through a one or more sections of the honeycomb structure **102**.

(22) In order to maintain a generally uniform cell density (e.g., number of cells per square inch) between, for example, radially inner and outer portions of the honeycomb structure **102**, the number of radial walls **108** is varied at different radial positions. In some embodiments, the honeycomb structure comprises cell transitions **124** located at selected points along the radius of the honeycomb structure **102**, with respect to which the number of radial walls is decreased at radial distances closer to the central axis C. At these cell transitions **124**, the number of radial walls **108**, and therefore the number of cells per unit length of the angular walls, is different on opposite radial sides of the cell transition. For example, the number of cells can be reduced by a designated amount, such as by $\frac{1}{2}$, $\frac{1}{4}$, or $\frac{1}{8}$ (e.g., or doubled, tripled, quadrupled) or any other suitable amount. FIG. 1 illustrates an embodiment in which the number of cells **112**, is halved at each cell transition **124** moving radially inwardly. For example, the number of cells **112** on the radially inward side of cell transition **124a** in the octant of FIG. 1 is 2, while the number of cells **112** on the radially outward side of the cell transition **124a** in the octant of FIG. 1 is 4. Likewise, moving radially outwardly, the number of cells **112** doubles from 4 to 8 on opposite sides of the cell transition **124b** in the octant of FIG. 1. Each cell transition **124** can be defined by a particular one of the angular walls **110** located at each designated radial distance, which is referred to as a transitional angular wall **126** and emphasized for clarity in FIG. 1.

(23) The cell transitions **124** can divide the honeycomb structure **102** into a plurality of resistive regions **128**. Each resistive region **128** can comprise a plurality of cells **112** and be radially bounded on its sides by one of the cell transitions **124** and/or transitional angular walls, by the central electrode **104**, by the skin **114**, and/or by the peripheral electrode **106**. For example, resistive region **128a** is radially bounded on one side by cell transition **124a** and on another side by central electrode **104**. Similarly, resistive region **128d** is radially bounded on one side by cell

transition **124c** and on another side by cell transition **124d**. While four cell transitions and five resistive regions are depicted in FIG. **1**, electric heater **100** can comprise other numbers of cell transitions, dividing the honeycomb structure into a corresponding number of resistive regions **128**.

(24) In order to maintain uniformity of the cross sectional area of the cells **112** in each of the resistive regions **128**, the radial dimension (or radial length) of each cell (i.e., the distance between the radially adjacent angular walls **110** forming each cell) can change at different radial distances from the central axis C. For example, FIG. **1** shows radial dimensions D1 and D2 for two different cells **112** in the same resistive region but at different radial distances from the central axis C, which illustrates that the radial dimension decreases the closer in radial proximity to the skin **114**. Reducing the radial dimension of the cells **112** in this way can be used to compensate for the angular widening of the cells **112** caused by the angular divergence between angularly adjacent radial walls **108** as they radiate from the central axis C of the honeycomb structure **102**. In this way, the radial and angular dimensions of the cells can be varied in some embodiments in order to maintain a generally uniform hydraulic diameter of the cells **112** across the entire face of the honeycomb structure **102**. In some embodiments, the hydraulic diameter of the cells **112** is within $\pm 35\%$ variation of each other across the entire face of the honeycomb structure **102**.

(25) FIG. **2** depicts a perspective view of the electrical heater **100**, showing central axis C extending longitudinally through electrical heater **100**, and peripheral electrode **106** disposed on the outer skin **114** of the honeycomb structure **102**. While the example of FIGS. **1** and **2** are illustrated as having right-cylindrical outer shapes with a central axis C corresponding to an axis of rotation, in alternate examples, electrical heater **100** can be other shapes, such as those having an elliptical cross-section. In such embodiments, the honeycomb structure can be arranged with respect to multiple central axis, e.g., one for each focal point of an elliptical shape, with radial walls radiating from each focal point.

(26) Electric heater **100** is configured to heat a flow of gas as the gas travels from inlet face **130** to outlet face **132**. The honeycomb structure **102** can be coated with a catalyst to react with components of the gas (e.g., pollutants in an engine exhaust stream), and/or the catalyst can be included on a separate downstream treatment device (e.g., a honeycomb substrate loaded with the catalyst) that is heated by proximity to the heater **100** and by the flow of gas after the flow of gas is heated by the electric heater **100**. The peripheral electrode can cover the entirety or only portions of the circumference of outer skin **114**, as can be suitable for directing current to portions of the heater **100** and as will be discussed in more detail in connection with FIG. **14**.

(27) As shown in FIG. **2**, heater **100** can be disposed upstream of an exhaust treatment device **133**, such as a catalytic substrate, particulate filter, or partial filter. Similar to heater **100**, exhaust treatment device **133** can comprise a honeycomb body comprising an outer skin **135** surrounding a honeycomb structure **137**. For example, the exhaust treatment device **133** can be formed, for example, by extrusion of a ceramic forming material through an extrusion die as a green body, which is then cut, dried, and fired into a final ceramic honeycomb body. The ceramic forming material can comprise inorganics (e.g., alumina, silica, etc.), binders (e.g., methylcellulose), pore formers (e.g., starch, graphite, resins), a liquid vehicle (e.g., water), sintering aids, or any other additives helpful in the manufacture of the final ceramic honeycomb body. The final ceramic honeycomb body can comprise cordierite, aluminum titanate, alumina, mullite, silicon carbide, and/or other ceramic materials, or combinations thereof.

(28) The honeycomb structure **137** of the exhaust treatment device **133** can comprise a plurality of intersecting walls, forming a plurality of contiguous cells. At least a subset of the cells of the exhaust treatment device **133** extend axially between opposing end faces **139**, **141** of the exhaust treatment device **133** to define a set of channels through which a gas can flow. The inlet face **139**, can thus be in fluid communication with the outlet face **141** via the channels, so that a gas, incident upon inlet end face **139** flow through the channels formed by the cells and exhaust outlet face **141**.

(29) The cells of the exhaust treatment device **133**, in cross-section can be formed of any suitable

shape, including, for example, square, rectangular, triangular, or hexagonal. The cells of the exhaust treatment device **133** need not be the same shape or dimension as the cells **112** of the heater **100**. Indeed, the cells of the exhaust treatment device **133** can be larger or smaller, have a different cells per square inch (cps) than the cells **112** of the heater **100**, and/or can be a different shape in cross-section. The honeycomb structure **137** can be impregnated or coated with a catalyst, that, when heated by the heater **100** (e.g., by proximity to the heater **100** and/or indirectly by an exhaust stream that is heated by heater **100**), activates to react with the exhaust. In some embodiments, only the electric heater **100** is loaded with a catalytic material, only the exhaust treatment device is loaded with a catalytic material, or both the heater **100** and the exhaust treatment device **133** are loaded with catalytic materials.

(30) The current generated in heater **100** serves to heat the flow of gas, e.g., vehicle exhaust stream, which flows through exhaust treatment device **133**. Thus, the heater **100** heats the catalyst of the exhaust treatment device **133**. In some embodiments, the heater **100** is dimensioned relatively to exhaust treatment device **133**, so that heater **100** will heat relatively quickly in comparison to the exhaust treatment device **133**. For example, a heater used with an exhaust treatment device **133** having an axial length of 6 in., can have an axial length of between about a quarter of an inch and an inch. If the heater **100** and exhaust treatment device **133** are constructed from different materials and/or with different cell geometries, the heater **100** and exhaust treatment device **133** can be extruded as a separate honeycomb bodies. However, in some embodiments, the heater **100** and the exhaust treatment device **133** can be combined by extruding a single honeycomb body (e.g., of a conductive ceramic material) and applying the electrodes **104**, **106** directly to this single honeycomb body. In such an embodiment, the electrodes can be arranged to span only a portion of the axial length of this single honeycomb body.

(31) As mentioned above, a potential applied between central electrode **104** and peripheral electrode **106** of FIG. **1** causes current to flow radially from central electrode toward peripheral electrode **106** (in alternative examples, current can flow from peripheral electrode **106** toward central electrode **104** as a result of the opposite potential applied between the central electrode **104** and peripheral electrode **106**). However, at each cell transition **124**, the current will be divided between the additional parallel radial walls **108** disposed on the outer side of the cell transition **124**. If the number of radial walls **108** is doubled on the radially outer side of the cell transition **124** with respect to the inner side of the cell transition **124**, the equivalent resistance on the radially outer side of the cell transition **124** is one half the resistance of equivalent resistance on the radially inner side of the cell transition **124** (assuming all other aspects of the radial and angular walls are equal). The resulting drop in equivalent resistance will create a corresponding drop in generated heat on the radially outer side of the cell transition, resulting in an undesirable non-uniform heat profile in the resistive regions featuring higher numbers of parallel radial walls **108**.

(32) FIG. **3** schematically depicts the equivalent resistance of a radial wall **108a** of a radially inner resistive region at a cell transition (e.g., resistive region **128c** with respect to cell transition **124c**) to the radial walls **108b**, **108c** of a radially outer resistive region (e.g., resistive region **128d** with respect to cell transitions **124c**). As shown, in this example, the resistance R of each radial walls **108** is substantially equal. However, the current I traveling through each radial wall **108b**, **108c**, is half the current I traveling through radial wall **108a**, because the current is divided equally between radial walls **108b** and **108c**. Accordingly, the heat generated by radial walls **108b**, **108c** of the outer resistive region **128** will be one quarter the heat generated by radial wall **108a** of the inner resistive region. This can be demonstrated by the following equations:

$$(33) \quad Q_{\text{inner}} = (I)^2 * R \quad (1) \quad Q_{\text{outer}} = \left(\frac{I}{2}\right)^2 * R \quad (2)$$

where $Q_{\text{sub.inner}}$ is the heat generated by radial wall **108a** of the inner resistive region and $Q_{\text{sub.outer}}$ is the heat generate by each radial wall **108b**, **108c** of the outer resistive region (the equation for heat generation is the same as the equation for power dissipated in each radial wall

108).

(34) To mitigate the drop in resistance at each cell transition **124**, the resistive region **128** on the radially inner side of the cell transition **124** can be modified to generate equivalent heat to the resistive region on the radially outer side of the cell transition **124** by changing the resistance of the radial walls **108** of the inner resistive region **128** with respect to the outer resistive region **128**.

Examples are discussed with respect to FIGS. 4A-6.

(35) FIG. 4A illustrates a simplified, partial cross-section view of one of the cell transitions **124** and radial walls **108a-108d**, according to a first method for changing the resistance of the radial walls **108** of the inner resistive region **128** with respect to the outer resistive region **128**, for a given one of the cell transitions **124**. With respect to the portion of the cell transition **124** depicted in FIG. 4A, radial wall **108a** is disposed in the inner resistive region **128**, while radial walls **108b**, **108c**, and **108d** are disposed in the outer resistive region **128**. In this example, although three radial walls **108d**, **108c**, **108d**, are shown for one radial wall **108a**, this example doubles the number of radial walls **108d** on the outer side of each cell transition **124**.

(36) As shown in FIG. 4A, in order change the resistance of the radial walls **108**, the wall thickness on the inner side of the cell transition **124** is increased, such that the heat generation on either side of the cell transition is substantially similar. For example, if but for the increase in thickness of the radial wall **108** each radial wall **108** on the outer side of the cell transition **124** would generate one quarter the heat of the radial wall **108** on the inner side of the cell transition **124**, the thickness of the radial wall **108** of the resistive region **128** on the inner side of the cell transition **124** can be increased by fourfold with respect to the thickness of each radial wall on the outer side of the transition region **124**. This is shown in FIG. 4A, where the thickness of radial wall **108a**, being located on the inner side of the depicted cell transition **124** is represented as 4X and the thickness of radial walls **108b-108d**, being located on the outer side of cell transition **124** is represented as X. As shown in FIG. 4B radial wall **108a** due to its thickness will have a resistance of some value R, whereas radial walls **108** will have a resistance of 4R. As a result, the heat dissipated $Q_{\text{sub.inner}}$ by radial wall **108a**, located on the inner side of a cell transition **124**, will be given by equation (1). Whereas, because radial walls **108b-108c** will receive half the current flowing through radial walls **108**, the heat generated by each radial wall **108b-108c**, being located on the outer side of cell transition **124**, can be given by eq. (3):

$$(37) \quad Q_{\text{outer}} = \left(\frac{I}{2}\right)^2 * 4R = Q_{\text{inner}} \quad (3)$$

Accordingly, the heat generated at each radial wall **108** will be substantially similar, promoting a uniform heat profile throughout resistive regions **128**.

(38) The thickness required to achieve uniform heating will depend, in part, on the ratio of the number of radial walls **108** in the outer resistive region **128** to the inner resistive region **128**. Generally speaking, all other aspects of the radial walls **108** being equal, the relative thickness of the radial walls **108** of the inner resistive region **128** will be equivalent to the inverse of the square of the ratio of the number of radial walls **108** of the outer resistive region **128** to the number of radial walls **108** of the inner resistive region **128**, due the equation for heat generation by each radial wall **108**. Thus, if there are three times as many radial walls **108** on the outer side of cell transition **124** as radial walls **108** on the inner side of the cell transition **124**, the radial walls **108** on the inner side of the cell transition **124** should be nine times as thick as the radial walls **108** on the outer side of the cell transition **124**. More generally, as other aspects of the radial walls **108** can vary (e.g., length, axial length, material, etc.), the thickness of the relative thickness of the radial walls **108** of the inner resistive region **128** will be equivalent to the inverse of the square of the ratio of the current flowing through each radial wall **108** on the outer side of cell transition **124** to the current flowing through each radial wall **108** on the inner side of cell transition **124**. Thus, in the example of FIGS. 4A and 4B, because half the current is flowing through the radial walls **108b-108d**, radial wall **108a** is four times as thick as radials walls **108b-108d**.

(39) FIG. 5 illustrates a simplified, partial cross-section view of a cell transition **124** and radials walls **108a-108d** according to a second embodiment for changing the resistance of the radial walls **108** of the inner resistive region **128** with respect to the outer resistive region **128**, for a given cell transition **124**. According the second embodiment, the length of the radial walls **108** on the inner side of the cell transition **124** are shortened, such that the resistance of each radial wall **108a** on the inner of cell transition **124** is some fraction of the resistance each radial wall **108b-108d** on the outer side of cell transition **124**. This example modifies the lengths of the radial walls **108** to vary the resistances of each the resistance of each radial wall **108** will increase with length. For example, if the radial walls **108** are doubled at the cell transition **124** on the outer side, the radial wall **108** on the inner side of the cell transition **124** can be shortened to one quarter of the length of the radial walls **108** on the outer side of the cell transition **124**. Thus, as shown in FIG. 5, the radial wall **108a** on the inner side of the cell transition **124** is length L , and the radial walls **108** on the outer side of the cell transition are length $4L$. Thus, returning to the example of FIG. 1, if this embodiment were followed, the length of radial walls of resistive region **128c**, for example, would be modified to be four times as long as the radial walls **108** of resistive region **128d**. Like the example of FIG. 4B, the resistance of radial wall **108a** on the inner side of cell transition **124** will be of a resistance R , while the resistance of each radial wall **108b-108d** on the outer side of cell transition **124** will each be of a resistance $4R$. The heat generated by each resistive region **128** can, therefore, be calculated by Eq. (1) and Eq. (3), and will thus be equivalent. Regardless of the ratio of the number of radials walls **108** on the inner side of each cell transition **124** to the outer side of each cell transition **124**, the length of radial wall **108** on the inner side of cell transition **124** can be made equal to the inverse of the square of the ratio of current flowing through each radial wall **108** on the outer side of the cell transition **124** to the current flowing through each radial wall **108** on the inner side of cell transition **124**, in order to implement the second embodiment.

(40) In a third embodiment, as shown in FIG. 6 which is a cross-section taken in a plane along central axis C and bisecting the inlet face **130** of electric heater **100** the honeycomb is arranged in a generally conical shape, as opposed to a cylindrical shape shown in FIG. 2. For example, the axial length of the walls of the honeycomb body are increased toward the center axis C and decrease, progressively, toward the outer periphery of the honeycomb structure **102**. Thus, directly adjacent the central electrode **104**, the axial length of the honeycomb structure is longest, having a length represented as $L1$, while at the midpoint between the central electrode **104** and the outer periphery of the honeycomb structure, the axial length has diminished to $L2$. Finally, near to the outer periphery, the axial length has diminished to its shortest length, $L3$. Like the examples of FIG. 4 and FIG. 5, this example reduces the equivalent resistance of the honeycomb body toward the central axis C. In an alternative example, instead of reducing progressively, the axial length can change in steps, at, for example, each cell transition **124**. Thus, turning to FIG. 1, in an alternate example, the axial length of the walls in the resistive region **128c** can be made to be four times the length of the walls in the resistive region **128c**. As a result, like the example of FIGS. 4A-5, the resistance of radial wall **108** on the inner side of cell transition **124** will be resistance R , while the resistance of each radial wall on the outer side of cell transition **124** will be resistance $4R$. The heat generated by each radial can, therefore, be calculated by Eq. (1) for the radial walls **108** disposed in the inner resistive region **128** and Eq. (3) for the radial walls **108** disposed in the outer resistive region **128**, and will thus be equivalent. In the example in which the axial length is varied at each resistive region **128**, the relative axial length for each resistive region **128** can be similarly calculated as the inverse of the square of the ratio of the current flowing through each radial wall **108** on the outer side of cell transition **124** to the current flowing through each radial wall **108** on the inner side of cell transition **124**.

(41) Any of the embodiments described in connection with FIGS. 4A-6 can be implemented in any suitable combination. For example, rather than modifying resistive region **128c** to be four times as thick or a quarter the length of resistive region **128d**, resistive region **128c** can instead be made to

be twice as thick and half the length of resistive region **128d**, yielding the same equivalent resistance as the individual examples of FIGS. **4A-6**. Thus, the example of FIGS. **4A-4B** can be combined with the example of FIG. **5** and/or the example of FIG. **6**. Likewise, the example of FIG. **5** can be combined with the example of FIG. **6**.

(42) Furthermore, the thicknesses, lengths, and/or axial lengths of each resistive region **128** can be varied within a particular resistive region **128**, to achieve a desired heat profile. For example, if a particular area of a resistive region **128** receives less exhaust than another area e.g., an area near the periphery of heater **100** the thickness of the radial walls **108** located within that area can be made thinner, in order to increase the heat generated at that particular area in relation to the remainder of the resistive region **128** and/or honeycomb structure **102**. One of ordinary skill in the art, in conjunction with a review of this disclosure, will understand how such modifications may be made to account for non-uniform exhaust flow or other factors outside of the structure of the heater **100** that may influence the heat profile of heater **100**.

(43) In some embodiments, such as shown in FIG. **7**, the cell transitions **124** are removed in order to eliminate the different resistive regions **128**, and thus potentially simplify the manufacturing process and/or improve the heating profile. In order to maintain constant hydraulic diameter, the radial distance between radially adjacent angular walls **110** can be, moving from the central axis **C** toward the outer periphery, progressively narrowed, such that the spacing between angular walls **110** decreases the closer in radial proximity to outer skin **114**. This is shown in FIG. **7**, where the radial distance **X1** between central axis **C** and the first angular wall **110a** is greater than the radial distance **X2** between the first angular wall **110a** and the next radially adjacent angular wall **110b**. The radial distance between each successive angular wall **110** can be reduced such that the following expression is true: $X_n < X_{n-1} < \dots < X_2 < X_1$, and such that the hydraulic diameter of each cell is kept substantially similar ($\pm 35\%$ variation) as the distance between adjacent radial walls increases toward the outer periphery. As a result, and as shown in FIG. **7**, honeycomb structure **102** comprises a higher number of angular walls **110** at the outer periphery than near the central axis **C**.

(44) An example of a thermal output that can be produced by the embodiment of FIG. **7** is illustrated in FIG. **8**. Because the radial walls **108** represent the shortest path between the electrodes **104**, **106**, and thus carry the bulk of the current, the current density near the electrode **104** is greater than the current density at the outer periphery, as the radial walls **108** are spaced relatively close together near the central axis **C**, but far apart near the outer periphery due to their radial divergence. As a result, this embodiment is most beneficial in situations in which excess heat generation is desired near the center. For example, FIG. **8** depicts an example heat profile for a heater resembling the embodiment of FIG. **7**. As shown, the thermal profile of FIG. **8** has a temperature of 860 K near the central electrode, 560 K at a midpoint between the central electrode and the outer periphery, and 430 K near the outer periphery, yielding a temperature differential of 430 K. The temperature values are provided for sake of discussion only and other temperatures can be achieved based on the particular material, geometry, voltage, and other parameters or characteristics of the heater.

(45) To improve temperature uniformity, the radial walls **108** can be staggered. For example, the radial walls **108** can be staggered at the intersections with each angular wall **110**, such that radially adjacent ones of the cells **112** of honeycomb structure **102** are offset, arranging the cells **112** in a brick-like pattern, as shown in FIG. **9**. Thus, as shown in FIG. **10** each radial wall **108** can be angularly offset from the radial walls **108** of radially adjacent cells by an angle θ_0 having a vertex at central axis **C**. For example, radial wall **108e** is offset by the angle θ from radial walls **108f** and/or **108g** because cell **112a**, partially defined by radial wall **108e** is radially adjacent to cell **112b**, which is partially defined by radial walls **108f** and **108g**. Due to the radial adjacency of the cells **112a** and **112b**, each of the radial walls **108e**, **108f**, and **108g** is commonly connected by the angular wall **110c**, which angular wall **110c** is commonly shared by the cells **112a** and **112b**.

(46) The architecture of FIG. **9** eliminates any straight path for current to flow from the central electrode **104** to the peripheral electrode **106**, forcing current to flow along the angular walls **110**,

thus causing more uniform heating throughout the heater, particularly near the outer periphery where the relative density of angular walls is greater. FIG. 11 shows an example of a temperature profile of a staggered honeycomb structure, e.g., that of FIG. 9, showing an improvement in the uniformity of the heat profile compared to the temperature profile of FIG. 8. Indeed, as shown, the structure of FIG. 11 has a temperature of 680 K near the central electrode, 520 K at a midpoint between the central electrode and the outer periphery, and 400 K near the outer periphery, yielding a temperature differential of 280 K.

(47) The distance the current flows along the angular wall 110 will be determined, at least in part, by the angle between a given radial wall 108 and the nearest radial walls 108 that define the radially adjacent cells. That is, current will primarily flow along the portion of the angular wall 110 separating the radial walls 108, in connecting the shortest path between the electrodes 104, 106. Thus, if the two radial walls 108 are separated by only a small degree, the current will only flow along angular wall 110 over the short angular distance between the radial walls 108. By contrast, if the radial walls 108 are offset by a distance equivalent to half the width of a cell, (such as shown in FIGS. 9 and 10) each radial wall will be equidistant from two radial walls 108 opposite a single angular wall 110, and, thus, equal current will flow from both radial walls 108, along the entire angular wall 110 (all other aspects of the cells 112 being equal). Accordingly, the amount of current flowing and the heat generated along the angular walls 110 can be adjusted, by varying the offset of the radial walls 108.

(48) The example of FIGS. 9 and 10, and the resulting heat profile of FIG. 11, can still feature greater heat generation near the central axis C than required in some embodiments. That is, for example, even more uniform temperature profiles may be beneficial for heating flows of gas that are uniform in flowrate across the face of the heater 100. Thus, to further increase the uniformity of the temperature profile, the radial walls near the center of the heater 100 can be aligned with the adjacent radial walls to prevent current flow in the angular walls near the core, whereas the radial walls 108, and, accordingly, the cells 112, near the outer periphery can be offset. Thus, as shown in the example of FIG. 12, the cells in a first region 105a resemble the embodiment of FIG. 7 in which the radial walls 108 propagate past multiple cells 112 in straight radial line, while the radial walls 108 in a second region 105b resemble that of the embodiment of FIG. 9, in which the radial walls 108 are staggered at each intersection of the radial walls 108 with the angular walls 110.

(49) The offset between radially adjacent cells 112 can be varied to achieve a desired heat profile. For example, if a particular area of heater 100 receives less exhaust than another area e.g., an area near the periphery of heater 100 the offset between radially adjacent cells 112 can be increased to force more current to flow thorough the angular walls 110 in a particular area. One of ordinary skill in the art, in conjunction with a review of this disclosure, will understand how such modifications may be made to account for non-uniform exhaust flow or other factors outside of the structure of the heater 100 that may influence the heat profile of 100.

(50) In some embodiments, particularly those in which the flow of gas is expected to be non-uniform across the inlet face of the heater 100, the cell dimensions within a particular area can be set to compensate for increased exhaust flow in certain areas. For example, the central area (i.e., the area disposed directly adjacent the central axis C, such as region 105a in FIG. 12) may receive increased exhaust flow with respect to the exhaust flow received at the outer periphery (e.g., as a result of the design of the exhaust treatment system installed on a vehicle, for example).

Accordingly, the cells 112 in this example can be divided into a first group that is disposed in the region directly adjacent the central axis (e.g., the region 105a in FIG. 12), and a second group that is disposed in a second region annularly arranged about the first group (e.g., the region 105b in FIG. 12). The hydraulic diameters of the cells 112 of the first group can be smaller or larger than the hydraulic diameters of the cells 112 of the second group. For example, by reducing the hydraulic diameter in the group of cells in the first radially inner region, the surface area of the walls, and thus the amount of catalytic material, exposed to the flow at the center of the heater can

be increased to maximize the catalytic effect in this area. Advantageously, this can be utilized to provide an increased temperature and/or catalytic loading in areas of higher gas flowrate. In other examples, the hydraulic diameter of other regions can be accordingly increased or decreased to accommodate the particular flow profile experienced, or expected to be experienced, by the heater **100** during use.

(51) In other examples, the modifications described in connection with FIGS. **4A-6** can be combined with the design(s) of FIGS. **9-12**, in various possible ways, to result in a more uniform heat profile. For example, the radial wall thickness of the design of FIG. **4** can be increased, such that the radial walls **108**, in general, are thicker (e.g., twice as thick) as the angular walls **110**, resulting in greater heating at the angular walls (e.g., by a factor of two). Thus, at the outer periphery, where the concentration of angular walls **110** is higher, the heat generation is greater, resulting in the heat profile of FIG. **13**, which is particularly uniform. As shown in FIG. **13**, a staggered cell design with increased radial wall thickness has a temperature of 52.0 K near the central electrode, 52.0 K at a midpoint between the central electrode and the outer periphery, and 500 K near the outer periphery, yielding a temperature differential of 20 K.

(52) FIG. **14** illustrates an embodiment that comprises a plurality of the peripheral electrodes **106**. For example, because current will flow from central electrode **104** to the peripheral electrode(s) **106**, the temperature profile can be adjusted by sizing and positioning the peripheral electrode(s) **106** where relatively higher temperatures are desired. For example, one or more peripheral electrodes **106** can be positioned to induce a current flow through cells **112** that are expected to receive greater rate of gas flow.

(53) While several inventive examples have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive examples described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the inventive teachings is/are used. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific inventive examples described herein. It is, therefore, to be understood that the foregoing examples are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive examples can be practiced otherwise than as specifically described and claimed. Inventive examples of the present disclosure are directed to each individual feature, system, article, material, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, and/or methods, if such features, systems, articles, materials, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure.

Claims

1. An electrical heater for heating an exhaust flow comprising: a honeycomb body comprising: a honeycomb structure comprising a central axis extending longitudinally therethrough and a plurality of interconnected walls, the interconnected walls comprising: a plurality of radial walls, each radial wall extending at least partially along a radius of the honeycomb body between the central axis and an outermost periphery of the honeycomb body; and a plurality of angular walls arranged concentrically with respect to the central axis and spanning between the radial walls; and a plurality of cells defined by the interconnected walls; a first electrode disposed at the central axis; and a second electrode disposed radially outwardly of the central axis and in electrical communication with the first electrode via one or more of the interconnected walls that are located

between the first electrode and second electrode, wherein the honeycomb body comprises a first resistive region separated from a second resistive region by a transitional angular wall, wherein the first resistive region comprises fewer radial walls than the second resistive region, the first resistive region being located nearer to the central axis than the second resistive region and both the first resistive region and the second resistive region being located between the first and second electrodes.

2. The electrical heater of claim 1, wherein a radial distance between radially consecutive angular walls of the plurality of angular walls decreases in a radial proximity to the outer periphery of the honeycomb body such that a hydraulic diameter of each cell is substantially similar.

3. The electrical heater of claim 1, wherein a first group of the plurality of cells is in a first region directly adjacent to the central axis, wherein a second group of the plurality of cells is disposed in a second region radially outwardly of the first region, wherein a first hydraulic diameter of each cell of the first set of the plurality of cells is less than a second hydraulic diameter of each cell of the second set of the plurality of cells.

4. The electrical heater of claim 1, wherein each radial wall of a first cell of the plurality of cells is angularly offset with respect to the radial walls of a second cell of the plurality of cells that is radially adjacent to the first cell.

5. The electrical heater of claim 1, wherein the first resistive region and second resistive region are bounded by the transitional angular wall and one of the first electrode, the second electrode, or a second transitional angular wall.

6. The electrical heater of claim 1, wherein the radial walls of the first resistive region are proportionally thicker than the radial walls of the second resistive region, such that, when carrying current between the first and second electrodes, heat generated by each radial wall of the first resistive region is substantially equivalent to heat generated by each radial wall of second resistive region.

7. The electrical heater of claim 1, wherein the radial walls of the first resistive region are proportionally shorter than the radial walls of the second resistive region, such that, when carrying current between the first and second electrodes, heat generated by each radial wall of the first resistive region is substantially equivalent to heat generated by each radial wall of second resistive region.

8. The electrical heater of claim 1, wherein the radial walls of the first resistive region are proportionally axially longer than the radial walls of the second resistive region, such that, when carrying current between the first and second electrodes, heat generated by each radial wall of the first resistive region is substantially equivalent to heat generated by each radial wall of second resistive region.

9. An electrical heater, for heating an exhaust flow comprising: a honeycomb body comprising: a honeycomb structure comprising a central axis extending longitudinally therethrough and a plurality of interconnected walls, the interconnected walls comprising: a plurality of radial walls, each radial wall extending at least partially along a radius of the honeycomb body between the central axis and an outermost periphery of the honeycomb body; and a plurality of angular walls arranged concentrically with respect to the central axis and spanning between the radial walls; and a plurality of cells defined by the interconnected walls; a first electrode disposed at the central axis; and a second electrode disposed radially outwardly of the central axis and in electrical communication with the first electrode via one or more of the interconnected walls that are located between the first electrode and second electrode, wherein the radial walls are thicker than the angular walls.

10. The electrical heater of claim 1, wherein the second electrode is disposed at an outer skin disposed about the outer periphery of the honeycomb structure.

11. An exhaust treatment system comprising the electrical heater of claim 1 and an exhaust treatment device is positioned downstream of the electrical heater.

12. The exhaust treatment system of claim 11, wherein the exhaust treatment device comprises a substrate loaded with a catalytic material, a particulate filter, or a combination.

13. A method for heating a catalytic material, comprising the steps of: applying an electric potential difference across a first electrode located at a central axis of a first honeycomb body and a second electrode located on the honeycomb body radially outwardly from the first electrode to heat an exhaust stream flowing through the first honeycomb body, wherein the honeycomb body comprises a plurality of intersecting walls comprising a plurality of radial walls, each radial wall extending at least partially along a radius of the honeycomb body between the central axis and an outermost periphery of the honeycomb body, at least some of the plurality of radial walls angularly spaced from each other about the central axis and a plurality of angular walls arranged concentrically with respect to the central axis and spanning between the radial walls, wherein the electric potential difference creates a flow of current through the intersecting walls located between the first electrode and the second electrode that generates resistive heating in the intersecting walls; and directing the exhaust stream to a second honeycomb body downstream of the first honeycomb body, wherein the second honeycomb body comprises the catalytic material, wherein the honeycomb body comprises a first resistive region separated from a second resistive region by a transitional angular wall, wherein the first resistive region comprises fewer radial walls than the second resistive region, the first resistive region being located nearer to the central axis than the second resistive region and both the first resistive region and the second resistive region being located between the first and second electrodes.

14. The method of claim 13, wherein the radial walls of the first resistive region are thicker than the radial walls of the second resistive region, such that the equivalent resistance of the first resistive region and the second resistive region are substantially similar.

15. The method of claim 13, wherein the radial walls of the first resistive region are shorter than the radial walls of the second resistive region, such that, when carrying current between the first and second electrodes, the heat generated by each radial wall of the first resistive region is substantially equivalent to the heat generated by each radial wall of the second resistive region.

16. The method of claim 13, wherein the radial walls of the first resistive region are axially longer than the radial walls of the second resistive region, such that, when carrying current between the first and second electrodes, the heat generated by each radial wall of the first resistive region is substantially equivalent to the heat generated by each radial wall of second resistive region.

17. The method of claim 13, wherein the radial walls are thicker than the angular walls.

18. The method of claim 13, wherein the second electrode is disposed at an outer skin disposed about the outer periphery.
